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CAMERA OPTICAL LENS

Abstract

Disclosed is a camera optical lens. The camera optical lens includes from an object side to an image side in sequence: a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. The camera optical lens satisfies the following conditions: $1.00 \leq d3/d1 \leq 3.00$; $0.20 \leq R3/R4 \leq 0.90$; $1.00 \leq f4/f3 \leq 4.00$; $0.50 \leq f6/f7 \leq 1.40$; $1.80 \leq nd8 \leq 2.20$. The thicknesses on-axis of the first lens and the second lens are $d1$ and $d3$, respectively. The curvature radius of the object side surface and image side surface of the second lens are $R3$ and $R4$, respectively. The focal lengths of the third lens, the fourth lens, the sixth lens, and the seventh lens are $f3$, $f4$, $f6$, and $f7$, respectively. The refractive index of the eighth lens is $nd8$. The camera optical lens has good optical performance and is characterized by a large aperture and miniaturization.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to Chinese Patent Application No. The present application is a continuation of PCT Patent Application No. PCT/CN2024/080087, entitled “CAMERA OPTICAL LENS,” filed Mar. 5, 2024, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of optical technology, in particular to a camera optical lens.

BACKGROUND

[0003] LIDAR (Light detection and ranging) is a radar system that detects the position, velocity, and other characteristics of a target by transmitting a laser beam. It is a system that integrates laser, Global Positioning system (GPS), and inertial navigation system (INS) technologies for acquiring data and generating an accurate digital elevation model (DEM) of the ground. The basic working principle of LIDAR is as follows: the detected signal (laser beam) is transmitted to the target, then the received signal reflected from the target (target echo) is compared with the transmitted signal, and after appropriate processing, relevant information about the target can be obtained, such as the target's distance, azimuth, altitude, speed, attitude, uniform shape, and other parameters, which can be used to detect, track, and identify the target, including aircraft and missiles. The sensor emits tens or hundreds of thousands of laser pulses per second, the timer starts when a light pulse is emitted and stops when the light pulse

(reflected from the first person/object) returns, the distance between the sensor and the person/object is calculated by measuring the time of flight (TOF) of the light pulse.

[0004] Although the existing lens mounted on the vehicle lidar has good optical performance, the setting of the focal length of the lens still has certain irrationality. As a result, the lens structure can not meet the design requirements of large aperture and miniaturization while having good optical performance.

SUMMARY

[0005] An object of embodiments of the present disclosure is to provide a camera optical lens, which can have good optical performance, and meet the design requirements for large aperture and miniaturization.

[0006] To resolve the foregoing technical problems, the present disclosure provides a camera optical lens, the camera optical lens from the object side to the image side in sequence includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. The camera optical lens satisfies the following conditions: $1.00 \leq d3/d1 \leq 3.00$; $0.20 \leq R3/R4 \leq 0.90$; $1.00 \leq f4/f3 \leq 4.00$; $0.50 \leq f6/f7 \leq 1.40$; $1.80 \leq nd8 \leq 2.20$. d3 represents a thickness on-axis of the first lens. d3 represents a thickness on-axis of the second lens. R3 represents a curvature radius of the object side surface of the second lens. R4 represents a curvature radius of the image side surface of the second lens. f3 represents a focal length of the third lens. f4 represents a focal length of the fourth lens. f6 represents a focal length of the sixth lens. f7 represents a focal length of the seventh lens. nd8 represents a refractive index of the eighth lens.

[0007] As an improvement, wherein the camera optical lens further satisfies the following conditions: $|f8/f| \leq 3.00$; where, f8 represents a focal length of the eighth lens; f represents a focal length of the camera optical lens.

[0008] As an improvement, wherein the camera optical lens has a field of view; and the camera optical lens further satisfies the following conditions: $(FOV \cdot f)/IH \geq 95$; where, FOV represents the field of view of the camera optical lens; f represents a focal length of the camera optical lens; IH represents an image height of the camera optical lens.

[0009] As an improvement, wherein the camera optical lens further satisfies the following conditions: $(FOV \cdot f)/IH \leq 125$.

[0010] As an improvement, wherein the first lens has a negative refractive power, and the first lens has a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $0.44 \leq (R1+R2)/(R1-R2) \leq 5.87$; $-7.27 \leq f1/f \leq -0.83$; $0.01 \leq d1/TTL \leq 0.05$; where, R1 represents a curvature radius of the object side surface of the first lens; R2 represents a curvature radius of the image side surface of the first lens; f1 represents a focal length of the first lens; f represents a focal length of the camera optical lens; TTL represents a total optical length of the camera optical lens.

[0011] As an improvement, wherein the second lens has a positive refractive power, and the second lens has a convex object side surface at a proximal axis and a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-34.47 \leq (R3+R4)/(R3-R4) \leq -1.01$; $3.18 \leq f2/f \leq 17.23$; $0.01 \leq d3/TTL \leq 0.08$; where, f2 represents a focal length of the second lens; f represents a focal length of the camera optical lens; TTL represents a total optical length of the camera optical lens.

[0012] As an improvement, wherein the third lens has a positive refractive power, and the third lens has a convex object side surface at a proximal axis and a convex image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $0.01 \leq (R5+R6)/(R5-R6) \leq 0.62$; $0.47 \leq f3/f \leq 2.47$; $0.03 \leq d5/TTL \leq 0.22$; where, R5 represents a curvature radius of the object side surface of the third lens; R6 represents a curvature radius of the image side surface of the third lens; f represents a focal length of the camera optical lens; d5 represents a thickness on-axis of the third lens; TTL represents a total optical length of the camera optical lens.

[0013] As an improvement, wherein the fourth lens has a positive refractive power, and the fourth lens has a convex object side surface at a proximal axis and a convex image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-1.98 \leq (R7+R8)/(R7-R8) \leq 1.12$; $0.76 \leq f4/f \leq 5.61$; $0.01 \leq d7/TTL \leq 0.23$; where, R7 represents a curvature radius of the object side surface of the fourth lens; R8 represents a curvature radius of the image side surface of the fourth lens; f represents a focal length of the camera optical lens; d7 represents a thickness on-axis of the fourth lens; TTL represents a total optical length of the camera optical lens.

[0014] As an improvement, wherein the fifth lens has a negative refractive power, and the fifth lens has a concave object side surface at a proximal axis and a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-0.99 \leq (R9+R10)/(R9-R10) \leq 1.45$; $-3.48 \leq f5/f \leq -0.45$; $0.00 \leq d9/TTL \leq 0.28$; where, R9 represents a curvature radius of the object side surface of the fifth lens; R10 represents a curvature radius of the image side surface of the fifth lens; f represents a focal length of the camera optical lens; d9 represents a thickness on-axis of the fifth lens; TTL represents a total optical length of the camera optical lens.

[0015] As an improvement, wherein the sixth lens has a positive refractive power, and the sixth lens has a convex object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-2.00 \leq (R11+R12)/(R11-R12) \leq 0.22$; $1.00 \leq f6/f \leq 5.66$; $0.01 \leq d11/TTL \leq 0.14$; where, R11 represents a curvature radius of the object side surface of the sixth lens; R12 represents a curvature radius of the image side surface of the sixth lens; f represents a focal length of the camera optical lens; d11 represents a thickness on-axis of the sixth lens; TTL represents a total optical length of the camera optical lens.

[0016] As an improvement, wherein the seventh lens has a positive refractive power, and the seventh lens has a convex object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-11.16 \leq (R13+R14)/(R13-R14) \leq -0.01$; $0.89 \leq f7/f \leq 10.88$; $0.01 \leq d13/TTL \leq 0.30$; where, R13 represents a curvature radius of the object side surface of the seventh lens; R14 represents a curvature radius of the image side surface of the seventh lens; f represents a focal length of the camera optical lens; d13 represents a thickness on-axis of the seventh lens; TTL represents a total optical length of the camera optical lens.

[0017] As an improvement, wherein the eighth lens has a negative refractive power, and the eighth lens has a concave object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-12.50 \leq (R15+R16)/(R15-R16) \leq 0.02$; $0.01 \leq d15/TTL \leq 0.05$; where, R15 represents a curvature radius of the object side surface of the eighth lens; R16 represents a curvature radius of the image side surface of the eighth lens; d15 represents a thickness on-axis

of the eighth lens; TTL represents a total optical length of the camera optical lens.
[0018] As an improvement, wherein the first lens is made of glass material.
[0019] As an improvement, wherein the second lens is made of glass material.
[0020] As an improvement, wherein the third lens is made of glass material.
[0021] As an improvement, wherein the fourth lens is made of glass material.
[0022] As an improvement, wherein the fifth lens is made of glass material.
[0023] As an improvement, wherein the sixth lens is made of glass material.
[0024] As an improvement, wherein the seventh lens is made of glass material.
[0025] As an improvement, wherein the eighth lens is made of glass material.
[0026] The beneficial effect of the present disclosure are as follows. The camera optical lens designed according to the present disclosure has excellent optical characteristics, and the camera optical lens can meet the design requirements for large aperture and miniaturization. The camera optical lens is particularly suitable for vehicle-mounted Lidar, cellular phone camera lens assemblies and WEB camera lenses, which includes camera elements such as CCD (Charge-Coupled Device), CMOS (Complementary Metal-Oxide-Semiconductor) and other camera elements for high pixels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] In order to more clearly illustrate the technical solutions in the embodiments of the present disclosure, the accompanying drawings to be used in the description in the embodiments will be briefly introduced hereinafter, and the following is a brief introduction of the drawings required in the description of the embodiments. It is obvious that the accompanying drawings in the description hereinafter are only some of the embodiments of the present disclosure, and that for a person having ordinary skill in the art, other accompanying drawings can also be obtained according to these drawings without creative labor.
[0028] FIG. 1 is a schematic diagram of a camera optical lens in accordance with a first embodiment of the present disclosure;
[0029] FIG. 2 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 1;
[0030] FIG. 3 shows the lateral color of the camera optical lens shown in FIG. 1;
[0031] FIG. 4 shows the longitudinal aberration of the camera optical lens shown in FIG. 1;
[0032] FIG. 5 is a schematic diagram of a camera optical lens in accordance with a second embodiment of the present disclosure;
[0033] FIG. 6 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 5;
[0034] FIG. 7 shows the lateral color of the camera optical lens shown in FIG. 5;
[0035] FIG. 8 shows the longitudinal aberration of the camera optical lens shown in FIG. 5;
[0036] FIG. 9 is a schematic diagram of a camera optical lens in accordance with a third embodiment of the present disclosure;
[0037] FIG. 10 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 9;
[0038] FIG. 11 shows the lateral color of the camera optical lens shown in FIG. 9;
[0039] FIG. 12 shows the longitudinal aberration of the camera optical lens shown in FIG. 9;
[0040] FIG. 13 is a schematic diagram of a camera optical lens in accordance with a fourth embodiment of the present disclosure;
[0041] FIG. 14 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 13;
[0042] FIG. 15 shows the lateral color of the camera optical lens shown in FIG. 13;
[0043] FIG. 16 shows the longitudinal aberration of the camera optical lens shown in FIG. 13;
[0044] FIG. 17 is a schematic diagram of a camera optical lens in accordance with a comparative embodiment of the present disclosure;
[0045] FIG. 18 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 17;
[0046] FIG. 19 shows the lateral color of the camera optical lens shown in FIG. 17;
[0047] FIG. 20 shows the longitudinal aberration of the camera optical lens shown in FIG. 17.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0048] To make the technical problems to be solved, technical solutions and beneficial effects of the present disclosure more apparent, the present disclosure is described in further detail together with the figure and the embodiments. It should be understood the specific embodiments described hereby is only to explain the present disclosure, not intended to limit the disclosure. It is understandable to a person having ordinary skill in the art that, in various embodiments of the disclosure, many technical details are proposed to enable the reader to better understand the present disclosure. However, even without the technical details and various variations and modifications based on the following embodiments, the technical solution claimed to be protected by the present disclosure can be realized.
[0049] In embodiments of the present disclosure, the terms “up,” “down,” “left,” “right,” “before,” “after,” “top,” “bottom,” “inside,” “outside,” “in,” “vertical,” “level,” “horizontal,” “vertical,” and the like indicate location or position relations based on the location or position relations shown in the accompanying drawings. These terms are intended primarily to better describe the present disclosure and its embodiments. These terms are not intended to qualify that the indicated device, elements, or components must have a particular orientation, or be constructed and operated in a particular orientation.
[0050] Besides, some of the above terms may be used to indicate other meanings besides orientation or position relations. For example, the term “up,” may also be used to indicate a certain attachment or connection relationship in some cases. To a person having ordinary skill in the art, the specific meaning of these terms in the present disclosure may be understood according to the specific circumstances.
[0051] In addition, the terms “install,” “set up,” “set up,” “open,” “connect,” “connect,” should be understood in a broad sense. For example, it may be a fixed connection, a removable connection, or a monolithic construction. It may be a mechanical connection or an electrical connection. It may be a direct connection, an indirect connection through an intermediate medium, or an internal connection between two devices, elements, or components. To a person having ordinary skill in the art, the specific

meaning of the terms in the present disclosure may be understood according to the specific circumstances.

[0052] It should be noted that in present disclosure, relational terms such as first and second are used only to distinguish one entity or operation from another and do not necessarily require or imply any such actual relationship or order between those entities or operations. Furthermore, the term “including,” “containing,” or any other variation thereof are intended to cover non-exclusive inclusion, such that a process, method, article, or apparatus comprising a set of elements includes not only those elements, but also other elements not expressly listed, or other elements as are inherent to such process, method, article or apparatus. Without further limitation, the fact that an element is defined by the phrase “including . . .” does not preclude the existence of additional identical elements in the process, method, article, or apparatus including the said element.

First Embodiment

[0053] Referring to FIG. 1, the first embodiment of the present disclosure provides a camera optical lens **10**. From the object side to the image side, the camera optical lens **10** includes in sequence: a first lens **L1**, a second lens **L2**, a third lens **L3**, a fourth lens **L4**, a fifth lens **L5**, a sixth lens **L6**, a seventh lens **L7**, and an eighth lens **L8**. The thickness on-axis of the first lens **L1** is defined as $d1$. The thickness on-axis of the second lens **L2** is defined as $d3$. The curvature radius of the object side surface of the second lens **L2** is defined as $R3$. The curvature radius of the image side surface of the second lens **L2** is defined as $R4$. The focal length of the third lens **L3** is defined as $f3$. The focal length of the fourth lens **L4** is defined as $f4$. The focal length of the sixth lens **L6** is defined as $f6$. The focal length of the seventh lens **L7** is defined as $f7$. The refractive index of the eighth lens **L8** is defined as $nd8$. And the following conditions should be satisfied:

$$1.00 \leq d3 / d1 \leq 3.00 \quad (1)$$

$$0.20 \leq R3 / R4 \leq 0.90 \quad (2)$$

[00001] $1.00 \leq f4 / f3 \leq 4.00 \quad (3)$

$$0.50 \leq f6 / f7 \leq 1.40 \quad (4)$$

$$1.80 \leq nd8 \leq 2.20 \quad (5)$$

[0054] In which, the formula (1) fixes the ratio of the thickness on-axis $d3$ of the second lens **L2** to the thickness on-axis $d1$ of the first lens **L1**. When the ratio is within the range fixed in the formula (1), it is beneficial for the compression of the total optical length TTL of the camera optical lens **10**, and it is beneficial for the realization of miniaturization.

[0055] The formula (2) fixes the shape of the second lens **L2**. When the formula (2) is satisfied, the degree of deflection of light passing through the second lens **L2** can be eased, and the lateral color can be effectively corrected, so that the lateral color is less than or equal to $6.0 \mu\text{m}$.

[0056] The formula (3) fixes the ratio of the focal length $f4$ of the fourth lens **L4** to the focal length $f3$ of the third lens **L3**. When the ratio is within the range fixed in the formula (3), the appropriate distribution of the optical focal length of the camera optical lens **10** results in that the camera optical lens **10** has better imaging quality and lower sensitivity.

[0057] The formula (4) fixes the ratio of the focal length $f6$ of the sixth lens **L6** to the focal length $f7$ of the seventh lens **L7**. It can be seen from the range limited by the formula (4) that the focal length $f6$ of the sixth lens **L6** and the focal length $f7$ of the seventh lens **L7** are relatively close to each other, which is beneficial for the light passing through the sixth lens **L6** and the seventh lens **L7** smoothly excessive, so that the image quality of the camera optical lens **10** can be improved.

[0058] The formula (5) fixes the refractive index of the eighth lens **L8**. It can be seen that the eighth lens **L8** is made of a material with a higher refractive index, so that it is beneficial to reduce the aperture of the rear end of the camera optical lens **10** and improve the image quality of the camera optical lens **10**.

[0059] In this embodiment, by setting a plurality of lenses (**L1**, **L2**, **L3**, **L4**, **L5**, **L6**, **L7**, **L8**), the ratio of the thickness of the second lens **L2** to the thickness of the first lens **L1**, the shape of the second lens **L2**, the ratio of the focal length $f3$ of the third lens **L3** to the focal length $f4$ of the fourth lens **L4**, the ratio of the focal length $f6$ of the sixth lens **L6** to the focal length $f7$ of the seventh lens **L7**, and the refractive index of the eighth lens **L8**, so that the camera optical lens **10** has good optical performance, and the camera optical lens has characteristics of large aperture and miniaturization. The camera optical lens **10** is particularly suitable for cellular phone camera lens assemblies, WEB camera lenses and vehicle-mounted Lidar lens, which includes camera elements such as CCD and CMOS for high pixel.

[0060] Preferably, the focal length of the eighth lens **L8** is defined as $f8$, the focal length of the camera optical lens **10** is defined as f , and the following condition should be satisfied:

[00002] $f8 / f \leq 3.00 \quad (6)$

[0061] The formula (6) fixes the focal length of the eighth lens **L8**. It can be seen that the focal length of the eighth lens **L8** is short, which is beneficial for improving the light receiving ability of the camera optical lens **10**, ensuring the luminous flux light flux and allowing for a smaller main ray angle (CRA, Chief Ray Angle), so that the relative illumination of the camera optical lens **10** is improved.

[0062] Preferably, the vision field angle of the camera optical lens **10** is defined as FOV (i.e., field of view), the focal length of the camera optical lens **10** is defined as f , the image height of the camera optical lens **10** is defined as IH . The following condition shall be satisfied:

[00003] $(FOV * f) / IH \geq 95 \quad (7)$

[0063] Within the range fixed in the formula (7), the camera optical lens **10** can be made to take into account both the field of view and long focus to realize medium and long-distance imaging.

[0064] Preferably, the camera optical lens **10** also may be satisfied with the following condition:

[00004] $(FOV * f) / IH \leq 125 \quad (8)$

[0065] In the embodiment, the first lens **L1** has a concave object side surface at the proximal axis and a concave image side surface at the proximal axis. The first lens **L1** has a negative refractive power. In other alternative embodiment, the first lens **L1** may also have a positive refractive power, and the object side surface and the image side surface of the first lens **L1** may also be

provided with other concave and convex distributions.

[0066] Preferably, the curvature radius of the object side surface of the first lens L1 is defined as R1, the curvature radius of the image side surface of the first lens L1 is defined as R2, the focal length of the first lens L1 is defined as f1, the focal length of the camera optical lens 10 is defined as f, and the total optical length of the camera optical lens 10 is defined as TTL. The following conditions shall be satisfied:

$$0.44 \leq (R1 + R2) / (R1 - R2) \leq 5.87 \quad (9)$$

[00005] $-7.27 \leq f1 / f \leq -0.83 \quad (10)$

$$0.01 \leq d1 / TTL \leq 0.05 \quad (11)$$

[0067] The formula (9) fixes the shape of the first lens L1. When the formula (9) is satisfied, it is beneficial for improving the imaging quality of the camera optical lens 10. Preferably, the following condition shall be satisfied, $0.70 \leq (R1+R2)/(R1-R2) \leq 4.70$. The formula (10) fixes the ratio of the focal length f1 of the first lens L1 to the focal length f of the camera optical lens 10. When the formula (10) is satisfied, it is beneficial for improving the optical performance of the camera optical lens. The following condition shall be satisfied, $-4.55 \leq f1/f \leq -1.04$. The formula (11) fixes the ratio of the thickness on-axis d1 of the first lens L1 to the total optical length TTL of the camera optical lens 10. When the formula (11) is satisfied, it is conducive to the realization of ultra-thin design. The following condition shall be satisfied, $0.01 \leq d1/TTL \leq 0.04$.

[0068] In this embodiment, the second lens L2 has a convex object side surface at the proximal axis and a concave image side surface at the proximal axis. The second lens L2 has a positive refractive power. In other alternative embodiments, the object side surface and the image side surface of the second lens L2 may also be provided with other concave and convex distributions, and the second lens L2 may also have a negative refractive power.

[0069] Preferably, the focal length of the second lens L2 is defined as f2, the focal length of the camera optical lens 10 is defined as f, and the total optical length of the camera optical lens 10 is defined as TTL. The following conditions should be satisfied:

$$-34.47 \leq (R3 + R4) / (R3 - R4) \leq -1.01 \quad (12)$$

[00006] $3.18 \leq f2 / f \leq 17.23 \quad (13)$

$$0.01 \leq d3 / TTL \leq 0.08 \quad (14)$$

[0070] The formula (12) fixes the shape of the second lens L2. When the formula (12) is satisfied, it is beneficial for correcting the aberration of the image on-axis and the like of the camera optical lens 10. Preferably, the following condition shall be satisfied, $-21.54 \leq (R3+R4)/(R3-R4) \leq -1.26$. The formula (13) fixes the ratio of the focal length f2 of the second lens L2 to the focal length f of the camera optical lens 10. When the formula (13) is satisfied, the second lens L2 has the appropriate positive refractive power, which is conducive to the design of ultra-thin and wide-angle of the camera optical lens. The following condition shall be satisfied, $5.09 \leq f2/f \leq 13.78$. The formula (14) fixes the ratio of the thickness on-axis d3 of the second lens L2 to the total optical length TTL of the camera optical lens 10. When the formula (14) is satisfied, it is conducive to the realization of ultra-thin design. Preferably, the following condition shall be satisfied, $0.02 \leq d3/TTL \leq 0.07$.

[0071] In this embodiment, the third lens L3 has a convex object side surface at the proximal axis and a convex image side surface at the proximal axis. The third lens L3 has a positive refractive power. In other alternative embodiments, the object side surface and the image side surface of the third lens L3 may also be provided with other concave and convex distributions, and the third lens L3 may also have a negative refractive power.

[0072] Preferably, the curvature radius of the object side surface of the third lens L3 is defined as R5, the curvature radius of the image side surface of the third lens L3 is defined as R6, the focal length of the camera optical lens 10 is defined as f, the thickness on-axis of the third lens L3 is defined as d5, the total optical length of the camera optical lens 10 is defined as TTL. The following conditions should be satisfied:

$$0.01 \leq (R5 + R6) / (R5 - R6) \leq 0.62 \quad (15)$$

[00007] $0.47 \leq f3 / f \leq 2.47 \quad (16)$

$$0.03 \leq d5 / TTL \leq 0.22 \quad (17)$$

[0073] The formula (15) fixes the shape of the third lens L3. When the formula (15) is satisfied, it is beneficial for improving the imaging quality. Preferably, the following condition shall be satisfied, $0.02 \leq (R5+R6)/(R5-R6) \leq 0.49$. The formula (16) fixes the ratio of the focal length f3 of the third lens L3 to the focal length f of the camera optical lens 10, by which, the aberration can be reduced. Preferably, the following condition shall be satisfied, $0.75 \leq f3/f \leq 1.97$. The formula (17) fixes the ratio of the thickness on-axis d5 of the third lens L3 to the total optical length TTL of the camera optical lens 10. When the formula (17) is satisfied, it is beneficial to meet the design of ultra-thin of the camera optical lens 10. The following condition shall be satisfied, $0.05 \leq d5/TTL \leq 0.17$.

[0074] In this embodiment, the fourth lens L4 has a convex object side surface at the proximal axis and a convex image side surface at the proximal axis. The fourth lens L4 has a positive refractive power. In other alternative embodiments, the object side surface and the image side surface of the fourth lens L4 may also be provided with other concave and convex distributions, and the fourth lens L4 can also have a negative refractive power.

[0075] Preferably, the curvature radius of the object side surface of the fourth lens L4 is defined as R7, the curvature radius of the image side surfaces of the fourth lens L4 is defined as R8, the focal length of the camera optical lens 10 is defined as f, the thickness on-axis of the fourth lens L4 is defined as d7, the total optical length of the camera optical lens 10 is defined as TTL.

The following conditions should be satisfied:

$$-1.98 \leq (R7 + R8) / (R7 - R8) \leq 1.12 \quad (18)$$

[00008] $0.76 \leq f4 / f \leq 5.61 \quad (19)$

$$0.01 \leq d7 / TTL \leq 0.23 \quad (20)$$

[0076] The formula (18) fixes the shape of the fourth lens L4. When the formula (18) is satisfied, it is beneficial for correcting the

aberration on-axis, with the development of ultra-thin and wide angle of the camera optical lens. Preferably, the following condition shall be satisfied, $-1.24 \leq (R7+R8)/(R7-R8) \leq 0.89$. The formula (19) fixes the ratio of the focal length f_4 of the fourth lens **L4** to the focal length f of the camera optical lens **10**. When the formula (19) is satisfied, it is beneficial for reducing the aberration and improving the imaging quality. Preferably, the following condition shall be satisfied, $1.22 \leq f_4/f \leq 4.49$. The formula (20) fixes the ratio of the thickness on-axis d_7 of the fourth lens **L4** to the total optical length TTL of the camera optical lens **10**, by which, it is beneficial for realizing the ultra-thin design of the camera optical lens. Preferably, the following condition shall be satisfied, $0.01 \leq d_7/TTL \leq 0.19$.

[0077] In this embodiment, the fifth lens **L5** has a concave object side surface at the proximal axis and a concave image side surface at the proximal axis. The fifth lens **L5** has a negative refractive power. In other alternative implementations, the object side surface and the image side surface of the fifth lens **L5** may also be provided with other concave and convex distributions, and the fifth lens **L5** may also have a positive refractive power.

[0078] Preferably, the curvature radius of the object side surface of the fifth lens **L5** is defined as R_9 , the curvature radius of the image side surface of the fifth lens **L5** is defined as R_{10} , the focal length of the fifth lens **L5** is defined as f_5 , the focal length of the camera optical lens **10** is defined as f , the thickness on-axis of the fifth lens **L5** is defined as d_9 , and the total optical length of the camera optical lens **10** is defined as TTL. The following conditions should be satisfied:

$$-0.99 \leq (R_9 + R_{10}) / (R_9 - R_{10}) \leq 1.45 \quad (21)$$

$$[00009] \quad -3.48 \leq f_5 / f \leq -0.45 \quad (22)$$

$$0.00 \leq d_9 / TTL \leq 0.28 \quad (23)$$

[0079] The formula (21) fixes the shape of the fifth lens **L5**, by which, it is beneficial for efficiently correcting the spherical aberration of the system. Preferably, the following condition shall be satisfied, $-0.62 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.16$. The formula (22) fixes the ratio of the focal length f_5 of the fifth lens **L5** to the focal length f of the camera optical lens **10**, by which, it is beneficial to improve the optical performance of the camera optical lens. Preferably, the following condition shall be satisfied, $-2.17 \leq f_5/f \leq -0.56$. The formula (23) fixes the ratio of the thickness on-axis d_9 of the fifth lens **L5** to the total optical length TTL of the camera optical lens **10**. When the formula (23) is satisfied, it is beneficial for the compression of the total optical length TTL of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $0.01 \leq d_9/TTL \leq 0.22$.

[0080] In this embodiment, the sixth lens **L6** has a convex object side surface at the proximal axis and a convex image side surface at the proximal axis, and the sixth lens **L6** has a positive refractive power. In other alternative embodiment, the sixth lens **L6** may also have a negative refractive power, and the object side surface and the image side surface of the sixth lens **L6** may also be provided with other concave and convex distributions.

[0081] Preferably, the curvature radius of the object side surface of the sixth lens **L6** is defined as R_{11} , the curvature radius of the image side surface of the sixth lens **L6** is defined as R_{12} , the focal length of the camera optical lens **10** is defined as f , the thickness on-axis of the sixth lens **L6** is defined as d_{11} , and the total optical length of the camera optical lens is defined as TTL. The following conditions should be satisfied:

$$-2.00 \leq (R_{11} + R_{12}) / (R_{11} - R_{12}) \leq 0.22 \quad (24)$$

$$[00010] \quad 1.00 \leq f_6 / f \leq 5.66 \quad (25)$$

$$0.01 \leq d_{11} / TTL \leq 0.14 \quad (26)$$

[0082] The formula (24) fixes the shape of the sixth lens **L6**, by which, the deflection degree of light passing through the lens can be eased, and the aberration can be effectively reduced. Preferably, the following condition shall be satisfied, $-1.25 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 0.17$. The formula (25) fixes the ratio of the focal length f_6 of the sixth lens **L6** to the focal length f of the camera optical lens **10**, by which, it is beneficial for reducing the aberration and improving the imaging quality. Preferably, the following condition shall be satisfied, $1.61 \leq f_6/f \leq 4.53$. The formula (26) fixes the ratio of the thickness on-axis d_{11} of the sixth lens **L6** to the total optical length TTL of the camera optical lens **10**, which is beneficial for compressing the total optical length TTL of the camera optical lens **10** and realizing the ultra-thin design of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $0.02 \leq d_{11}/TTL \leq 0.11$.

[0083] In this embodiment, the seventh lens **L7** has a convex object side surface at the proximal axis position and a convex image side surface at the proximal axis. The seventh lens **L7** has a positive refractive power. In other alternative embodiment, the seventh lens **L7** may also have a negative refractive power, the object side surface and the image side surface of the seventh lens **L7** may also be provided with other concave and convex distributions.

[0084] Preferably, the curvature radius of the object side surface of the seventh lens **L7** is defined as R_{13} , the curvature radius of the image side surface of the seventh lens **L7** is defined as R_{14} , the focal length of the camera optical lens **10** is defined as f , the thickness on-axis of the seventh lens **L7** is defined as d_{13} , and the total optical length of the camera optical lens **10** is defined as TTL. The following conditions should be satisfied:

$$-11.16 \leq (R_{13} + R_{14}) / (R_{13} - R_{14}) \leq -0.01 \quad (27)$$

$$[00011] \quad 0.89 \leq f_7 / f \leq 10.88 \quad (28)$$

$$0.01 \leq d_{13} / TTL \leq 0.30 \quad (29)$$

[0085] The formula (27) fixes the shape of the seventh lens **L7**, by which, it is beneficial for reducing the degree of deflection of light passing through the seventh lens **L7** and reducing the aberration effectively. Preferably, the following condition shall be satisfied, $-6.98 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq -0.01$. The formula (28) fixes the ratio of the focal length f_7 of the seventh lens **L7** to the focal length f of the camera optical lens **10**, by which, it is beneficial for improving the optical performance of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $1.43 \leq f_7/f \leq 8.71$. The formula (29) fixes the ratio of the thickness on-axis d_{13} of the seven lens **L7** to the total optical length TTL of the camera optical lens **10**. When the formula (29) is satisfied, it is beneficial for realizing the design of ultra-thin of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $0.02 \leq d_{13}/TTL \leq 0.24$.

[0086] In this embodiment, the eighth lens **L8** has a concave object side surface at the axis position and a concave image side surface at the axis position. The eighth lens **L8** has a negative refractive power. In other alternative embodiments, the eighth lens **L8** may also have a positive refractive power, and the object side surface and the image side surface of the eighth lens **L8** may also be provided with other concave and convex distributions.

[0087] Preferably, the curvature radius of the object side surface of the eighth lens **L8** is defined as R15, the curvature radius of the image side surface of the eighth lens **L8** is defined as R16, the thickness on-axis of the eighth lens **L8** is defined as d15, and the total optical length of the camera optical lens is defined as TTL. The following conditions should be satisfied:

[00012]
$$-12.5 \leq (R15 + R16) / (R15 - R16) \leq 0.02 \quad (30)$$

$$0.01 \leq d15 / TTL \leq 0.05 \quad (31)$$

[0088] The formula (30) fixes the shape of the eighth lens **L8**, by which, it is beneficial for correcting the aberration of the image on-axis, with the design of ultra-thin and wide angle. Preferably, the following condition shall be satisfied,

$-7.81 \leq (R15 + R16) / (R15 - R16) \leq 0.01$. The formula (31) fixes the ratio of the thickness on-axis d15 of the eighth lens **L8** to the total optical length TTL of the camera optical lens **10**, which is beneficial for control the total optical length TTL of the camera optical lens **10** reasonably. Preferably, the following condition shall be satisfied, $0.01 \leq d15 / TTL \leq 0.04$.

[0089] Preferably, the F number FNO of the camera optical lens **10** satisfies the following condition:

[00013]
$$FNO \leq 1.65 \quad (32)$$

[0090] The formula (32) fixes the F number of aperture of the camera optical lens **10**. When the formula (32) is satisfied, it is beneficial for the realization of miniaturization, and a larger amount of incoming light can be achieved.

[0091] In this embodiment, the first lens **L1**, the second lens **L2**, the third lens **L3**, the fourth lens **L4**, the fifth lens **L5**, the sixth lens **L6**, the seventh lens **L7** and the eighth lens **L8** may all be made of glass material. In other alternative embodiments, each lens may also be made of other materials.

[0092] Preferably, the first lens **L1** is a spherical lens, the second lens **L2** is a spherical lens, the third lens **L3** is an aspheric lens, the fourth lens **L4** is a spherical lens, the fifth lens **L5** is a spherical lens, the sixth lens **L6** is an aspheric lens, the seventh lens **L7** is a spherical lens, and the eighth lens **L8** is a spherical lens.

[0093] In this embodiment, optical elements like an optical filter GF may be arranged between the eighth lens **L8** and the image surface Si. The optical filter GF may be a glass cover plate or an optical filter sheet (filter). As shown in FIG. 1, an optical filter sheet GF is arranged between the eighth lens **L8** and the image surface Si. In other embodiments, the optical filter GF can also be arranged at other positions.

[0094] The camera optical lens **10** of the present disclosure has good optical performance and can meet the design requirements of a large aperture and minimization. The camera optical lens **10** is particularly suitable for cellular phone camera lens assemblies, WEB camera lenses and vehicle-mounted Lidar lens, which includes camera elements such as CCD and CMOS for high pixel.

[0095] The camera optical lens **10** of the present disclosure will be described below by way of examples. The various symbols recorded in each example are shown in table 1. The unit of the focal length, distance on-axis, curvature radius, thickness on-axis, location of inflection point, and location of stationary point is mm.

[0096] TTL: Total optical length (the distance on-axis from the object side surface of the first lens **L1** to the image Si), and the unit of TTL is mm.

[0097] Preferably, the camera optical lens **10** may also be provided with one or both of at least an inflection point and at least a stationary point on one or both of the object side surface and the image side surface, which is beneficial for meeting the requirements of high imaging quality. The specific embodiments are described below.

[0098] FIG. 1 is a schematic diagram of a camera optical lens in accordance with a first embodiment of the present disclosure. The design data of the camera optical lens **10** in the first embodiment of the present disclosure are described below.

[0099] Table 1 shows the curvature radius R, the thickness on-axis of the lens, the distance on-axis d between the lenses, the refractive index nd, and the Abbe number vd of the object side surface as well as the image side surface of the first lens **L1** to the eighth lens **L8** constituting the camera optical lens **10** in the first embodiment of the present disclosure. table 2 shows the conic coefficients k and the aspheric coefficients of the camera optic **10** as described. It is to be noted that in this embodiment, the unit of distance, radius and thickness is millimeter (mm).

TABLE-US-00001 TABLE 1 R d nd vd S1 ∞ d0= -4.690 R1 -1953.929 d1= 0.388 nd1 1.5891 vd1 61.25 R2 4.385 d2= 1.653 R3 29.476 d3= 1.162 nd2 1.7552 vd2 27.55 R4 72.748 d4= 3.768 R5 19.469 d5= 3.809 nd3 1.6192 vd3 63.85 R6 -8.143 d6= 0.703 R7 51.936 d7= 3.189 nd4 1.4970 vd4 81.59 R8 -10.890 d8= 0.107 R9 -14.525 d9= 0.542 nd5 1.7618 vd5 26.61 R10 17.706 d10= 0.541 R11 23.167 d11= 2.627 nd6 1.8513 vd6 40.10 R12 -17.279 d12= 0.116 R13 16.818 d13= 4.680 nd7 1.7340 vd7 51.49 R14 -17.208 d14= 0.108 R15 -16.743 d15= 0.915 nd8 1.8467 vd8 23.79 R16 16.359 d16= 1.873 R17 ∞ d17= 0.900 ndg 1.5168 vgd 64.21 R18 ∞ d18= 1.845

[0100] In which, the meaning of the various symbols is as follows.

[0101] R: The curvature radius of the optical surface, the central curvature radius in case of lens; [0102] S1: Aperture; [0103] R1: The curvature radius of the object side surface of the first lens **L1**; [0104] R2: The curvature radius of the image side surface of the first lens **L1**; [0105] R3: The curvature radius of the object side surface of the second lens **L2**; [0106] R4: The curvature radius of the image side surface of the second lens **L2**; [0107] R5: The curvature radius of the object side surface of the third lens **L3**; [0108] R6: The curvature radius of the image side surface of the third lens **L3**; [0109] R7: The curvature radius of the object side surface of the fourth lens **L4**; [0110] R8: The curvature radius of the image side surface of the fourth lens **L4**; [0111] R9: The curvature radius of the object side surface of the fifth lens **L5**; [0112] R10: The curvature radius of the image side surface of the fifth lens **L5**; [0113] R11: The curvature radius of the object side surface of the sixth lens **L6**; [0114] R12: The curvature radius of the image side surface of the sixth lens **L6**; [0115] R13: The curvature radius of the object side surface of the seventh lens **L7**; [0116] R14: The curvature radius of the image side surface of the seventh lens **L7**; [0117] R15: The curvature radius of the object side surface of the eighth lens **L8**; [0118] R16: The curvature radius of the image side surface of the eighth lens **L8**; [0119] R17: The curvature radius of the object side surface of the optical filter GF; [0120] R18: The curvature radius of the image side surface of

the optical filter GF; [0121] d: The thickness on-axis of the lens, or the distance on-axis between the lenses;
[0122] d0: The distance on-axis from the aperture S1 to the object side surface of the first lens L1;
[0123] d1: The thickness on-axis of the first lens L1;
[0124] d2: The distance on-axis from the image side surface of the first lens L1 to the object side surface of the second lens L2;
[0125] d3: The thickness on-axis of the second lens L2;
[0126] d4: The distance on-axis from the image side surface of the second lens L2 and the object side surface of the third lens L3;
[0127] d5: The thickness on-axis of the third lens L3;
[0128] d6: The distance on-axis from the image side surface of the third lens L3 to the object side surface of the fourth lens L4;
[0129] d7: The thickness on-axis of the fourth lens L4;
[0130] d8: The distance on-axis from the image side surface of the fourth lens L4 to the object side surface of the fifth lens L5;
[0131] d9: The thickness on-axis of the fifth lens L5;
[0132] d10: The distance on-axis from the image side surface of the fifth lens L5 to the object side surface of the sixth lens L6;
[0133] d11: The thickness on-axis of the sixth lens L6;
[0134] d12: The distance on-axis from the image side surface of the sixth lens L6 to the object side surface of the seventh lens L7;
[0135] d13: The thickness on-axis of the seventh lens L7;
[0136] d14: The distance on-axis from the image side surface of the seventh lens L7 to the object side surface of the eighth lens L8;
[0137] d15: The thickness on-axis of the eighth lens L8;
[0138] d16: The distance on-axis from the image side surface of the eighth lens L8 and the object side surface of the optical filter GF;
[0139] d17: The thickness on-axis of the optical filter GF;
[0140] d18: The distance on-axis from the image side surface of the optical filter GF and the image surface Si;
[0141] nd: The refractive power of d line (d line is green light with a wavelength of 550 nm);
[0142] nd1: The refractive power of the d line of the first lens L1;
[0143] nd2: The refractive power of the d line of the second lens L2;
[0144] nd3: The refractive power of the d line of the third lens L3;
[0145] nd4: The refractive power of the d line of the fourth lens L4;
[0146] nd5: The refractive power of the d line of the fifth lens L5;
[0147] nd6: The refractive power of the d line of the sixth lens L6;
[0148] nd7: The refractive power of the d line of the seventh lens L7;
[0149] nd8: The refractive power of the d line of the eighth lens L8;
[0150] ndg: The refractive power of d line of the optical filter GF;
[0151] vd: The abbe number;
[0152] vd1: The abbe number of the first lens L1;
[0153] vd 2: The abbe number of the second lens L2;
[0154] vd 3: The abbe number of the third lens L3;
[0155] vd 4: The abbe number of the fourth lens L4;
[0156] vd 5: The abbe number of the fifth lens L5;
[0157] vd 6: The abbe number of the sixth lens L6;
[0158] vd 7: The abbe number of the seventh lens L7;
[0159] vd 8: The abbe number of the eighth lens L8;
[0160] vg: The abbe number of the optical filter GF.

TABLE-US-00002 TABLE 2 conic coefficients aspheric coefficients k A4 A6 A8 A10 A12 R1 // // // R2 // // // R3 // // // R4 /
// // // R5 9.8980E+00 -5.2643E-04 -3.7881E-06 7.3542E-07 -6.3346E-08 2.2864E-09 R6 8.3998E-02 4.4298E-05
-3.8295E-06 6.1342E-07 -2.8578E-08 7.2265E-10 R7 // // // R8 // // // R9 // // // R10 // // // R11 9.2567E+00
-2.1889E-04 2.2568E-06 -1.8671E-07 5.6985E-09 -4.3425E-11 R12 -4.5949E+00 -2.2050E-05 -7.6684E-07
8.5361E-08 -2.6389E-09 6.6962E-11 R13 // // // R14 // // // R15 // // // R16 // // // conic coefficients aspheric coefficients k
A14 A16 A18 A20 / R1 // // // R2 // // // R3 // // // R4 // // // R5 9.8980E+00 -3.4393E-11 2.1856E-15 1.1945E-16
2.8708E-18 / R6 8.3998E-02 -6.4129E-12 -4.6093E-15 -1.9368E-16 -6.7051E-18 / R7 // // // R8 // // // R9 // // // R10 /
// // // R11 9.2567E+00 6.4076E-15 2.5096E-16 2.6253E-18 -1.3454E-19 / R12 -4.5949E+00 1.4775E-14 6.9632E-17
-2.4894E-18 -3.3290E-20 / R13 // // // R14 // // // R15 // // // R16 // // //

[0161] For convenience, the aspheric of each lens is used the aspherical surfaces shown in formula (33) below. However, the present disclosure is not limited to the aspheric polynomial form represented by the formula (33).

[00014]

$$z = (c^2 / r) / \{1 + [1 - (k + 1)(c^2 / r^2)]^{1/2}\} + A4c^4 + A6c^6 + A8c^8 + A10c^{10} + A12c^{12} + A14c^{14} + A16c^{16} + A18c^{18} + A20c^{20} \quad (33)$$

[0162] Where k is the cone coefficient, A4, A6, A8, A10, A12, A14, A16, A18 and A20 are aspheric coefficients, c is the curvature at the center of the optical surface, r is the vertical distance between the point on the aspheric curve and the optical axis, and z is the aspherical depth (i.e., the vertical distance between a point on the aspherical surface r from the optical axis and a section tangent to the vertex on the aspherical optical axis).

[0163] Table 3 and Table 4 respectively show the design data of the inflection point and stationary point of each lens of the camera optical lens 10 in the first embodiment of the present disclosure. Where P1R1 and P1R2 respectively represent the object side surface and the image side surface of the first lens L1, P2R1 and P2R2 respectively represent the object side surface and the image side surface of the second lens L2, P3R1 and P3R2 respectively represent the object side surface and the image side surface of the third lens L3, P4R1 and P4R2 respectively represent the object side surface and the image side surface of the fourth lens L4, P5R1 and P5R2 respectively represent the object side surface and the image side surface of the fifth lens L5, P6R1 and P6R2 respectively represent the object side surface and the image side surface of the sixth lens L6, P7R1 and P7R2 respectively

represent the object side surface and the image side surface of the seventh lens L7, and P8R1 and P8R2 respectively represent the object side surface and the image side surface of the eighth lens L8. The data corresponding to the field of “location of inflection point” is the vertical distance between the inflection point set on the surface of each lens and the optical axis of the camera optical lens 10. The data corresponding to the field of “location of stationary point” is the vertical distance from the stationary point set on the surface of each lens to the optical axis of the camera optical lens 10.

TABLE-US-00003 TABLE 3 The number of the location 1 of the location 2 of inflection points inflection point inflection point
P1R1 /// P1R2 /// P2R1 /// P2R2 /// P3R1 2 4.405 5.445 P3R2 /// P4R1 /// P4R2 /// P5R1 /// P5R2 /// P6R1 /// P6R2 1
4.915 / P7R1 /// P7R2 /// P8R1 /// P8R2 ///

TABLE-US-00004 TABLE 4 the number of stationary the location of stationary points point P1R1 // P1R2 // P2R1 // P2R2 //
P3R1 // P3R2 // P4R1 // P4R2 // P5R1 // P5R1 // P6R2 1 6.315 P6R1 // P7R1 // P7R2 // P8R1 // P8R2 //

[0164] Table 21, which appears later, shows the various values in the first embodiment, and the values corresponding with the parameters which are already fixed in the conditions.

[0165] FIG. 2 shows the schematic diagram of the field curvature and distortion after light with a wavelength of 555 nm passes through the camera optical lens 10 in the first embodiment. FIG. 3 shows the lateral color schematic diagram after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 10 in the first embodiment. FIG. 4 shows the longitudinal aberration after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 10 in the first embodiment.

[0166] As shown in Table 21, the first embodiment satisfies the various conditions.

[0167] In this embodiment, the pupil entering diameter (ENPD) of the camera optical lens 10 is 3.613 mm, the full vision field image height (IH) is 5.139 mm, and the vision field angle (FOV) in the diagonal direction is 107.40°. The camera optical lens 10 can meet the design requirements of large aperture and miniaturization, and chromatic aberrations on-axis and chromatic aberrations off-axis are adequately corrected. And the camera optical lens 10 has excellent optical characteristics.

Second Embodiment

[0168] FIG. 5 shows the camera optical lens 20 in the second embodiment of the present disclosure. The second embodiment is basically the same as the first embodiment, the meaning of its symbols is the same as that of the first embodiment, in the following, only the differences are described.

[0169] In the embodiment, the seventh lens L7 has a concave image side surface at the proximal axis, and the eighth lens L8 has a convex image side surface at the proximal axis.

[0170] Table 5 and table 6 show the design data of the camera optical lens 20 in the second embodiment of the present disclosure.

TABLE-US-00005 TABLE 5 R d nd vd S1 ∞ d0= -12.022 R1 -94.999 d1= 0.631 nd1 1.5891 vd1 61.25 R2 6.523 d2= 2.240 R3 29.728 d3= 1.890 nd2 1.7552 vd2 27.55 R4 145.061 d4= 7.056 R5 15.372 d5= 2.675 nd3 1.6192 vd3 63.85 R6 -10.519 d6= 0.200 R7 57.054 d7= 5.252 nd4 1.4970 vd4 81.59 R8 -8.393 d8= 0.050 R9 -11.409 d9= 0.250 nd5 1.7618 vd5 26.61 R10 18.613 d10= 0.165 R11 35.218 d11= 1.315 nd6 1.8513 vd6 40.10 R12 -60.657 d12= 1.050 R13 16.289 d13= 6.796 nd7 1.7340 vd7 51.49 R14 23.402 d14= 2.386 R15 -6.045 d15= 0.810 nd8 1.8467 vd8 23.79 R16 -8.347 d16= 0.200 R17 ∞ d17= 0.900 ndg 1.5168 vgd 64.21 R18 ∞ d18= 0.200

TABLE-US-00006 TABLE 6 conic coefficients aspheric coefficients k A4 A6 A8 A10 A12 R1 // // // R2 // // // R3 // // // R4 /
// // R5 8.1312E+00 -5.7574E-04 -2.1811E-05 6.0811E-07 -6.8337E-08 1.9317E-09 R6 1.8290E-01 1.0531E-04
-1.1082E-05 2.9700E-07 -2.7739E-08 8.6712E-10 R7 // // // R8 // // // R9 // // // R10 // // // R11 3.4105E+01
5.6290E-04 -3.9872E-06 -2.4854E-07 1.4357E-08 1.8238E-10 R12 -9.0058E+01 7.1455E-04 -2.9737E-06 1.3028E-07
4.2945E-09 1.1381E-10 R13 // // // R14 // // // R15 // // // R16 // // // conic coefficients aspheric coefficients k A14 A16 A18
A20 / R1 // // // R2 // // // R3 // // // R4 // // // R5 8.1312E+00 -4.2359E-11 2.4349E-13 1.3872E-14 -1.9386E-15 / R6
1.8290E-01 -7.8225E-12 -3.3555E-13 -1.1628E-14 3.3478E-16 / R7 // // // R8 // // // R9 // // // R10 // // // R11
3.4105E+01 -4.5014E-12 -3.8498E-13 -6.2337E-15 5.7074E-16 / R12 -9.0058E+01 -5.0029E-12 -2.0664E-13
-1.6728E-16 4.7221E-16 / R13 // // // R14 // // // R15 // // // R16 // // //

[0171] Table 7 and Table 8 respectively show the design data of the inflection point and stationary point of each lens of the camera optical lens 20 in the second embodiment of the present disclosure.

TABLE-US-00007 TABLE 7 The number of inflection the location 1 of inflection points point P1R1 // P1R2 // P2R1 // P2R2 //
P3R1 1 3.055 P3R2 // P4R1 // P4R2 // P5R1 // P5R2 // P6R1 // P6R2 1 1.355 P7R1 // P7R2 // P8R1 // P8R2 //

TABLE-US-00008 TABLE 8 the number of stationary points the location of stationary point P1R1 // P1R2 // P2R1 // P2R2 //
P3R1 1 4.415 P3R2 // P4R1 // P4R2 // P5R1 // P5R1 // P6R2 1 2.355 P6R1 // P7R1 // P7R2 // P8R1 // P8R2 //

[0172] Table 21, which appears later, shows the various values in the second embodiment, and the values corresponding with the parameters which are already fixed in the conditions.

[0173] FIG. 6 shows the schematic diagram of the field curvature and distortion after light with a wavelength of 555 nm passes through the camera optical lens 20 in the second embodiment. FIG. 7 shows the lateral color schematic diagram after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 20 in the second embodiment. FIG. 8 shows the longitudinal aberration after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 20 in the second embodiment.

[0174] As shown in Table 21, the second embodiment satisfies the various conditions.

[0175] In this embodiment, the pupil entering diameter (ENPD) of the camera optical lens 20 is 4.675 mm, the full vision field image height (IH) is 5.139 mm, and the vision field angle (FOV) in the diagonal direction is 77.60°. The camera optical lens 20 can meet the design requirements of large aperture and miniaturization, and chromatic aberrations on-axis and chromatic aberrations off-axis are adequately corrected. And the camera optical lens 20 has excellent optical characteristics.

Third Embodiment

[0176] FIG. 9 shows the camera optical lens 30 in the third embodiment of the present disclosure. The third embodiment is basically the same as the first embodiment, the meaning of its symbols is the same as that of the first embodiment, in the following, only the differences are described.

[0177] In this embodiment, the first lens of L1 has a convex object side surface at the proximal axis. The sixth lens L6 has a concave image side surface at the proximal axis. The seventh lens L7 has a concave image side surface at the proximal axis. The eighth lens L8 has a convex image side surface at the proximal axis.

[0178] Table 9 and table 10 show the design data of the camera optical lens 30 in the third embodiment of the present disclosure. TABLE-US-00009 TABLE 9 R d nd vd S1 ∞ d0= -12.898 R1 66.981 d1= 1.300 nd1 1.5891 vd1 61.25 R2 6.970 d2= 3.140 R3 10.852 d3= 1.315 nd2 1.7552 vd2 27.55 R4 12.189 d4= 6.754 R5 14.845 d5= 5.141 nd3 1.6192 vd3 63.85 R6 -13.070 d6= 0.200 R7 17.410 d7= 4.928 nd4 1.4970 vd4 81.59 R8 -8.378 d8= 0.050 R9 -10.583 d9= 0.250 nd5 1.7618 vd5 26.61 R10 31.535 d10= 1.441 R11 25.660 d11= 0.800 nd6 1.8513 vd6 40.10 R12 68311.732 d12= 1.686 R13 31.282 d13= 5.372 nd7 1.7340 vd7 51.49 R14 109.687 d14= 1.541 R15 -10.854 d15= 0.500 nd8 1.8052 vd8 25.48 R16 -281.556 d16= 0.199 R17 ∞ d17= 0.900 ndg 1.5168 vgd 64.21 R18 ∞ d18= 0.199

TABLE-US-00010 TABLE 10 conic coefficients aspheric coefficients k A4 A6 A8 A10 A12 R1 / / / / / / R2 / / / / / / R3 / / / / / / R4 / / / / / / R5 8.2388E+00 -4.7605E-04 -1.5285E-05 6.7999E-07 -6.7779E-08 2.0500E-09 R6 -1.5012E-01 3.0849E-05 -2.9007E-06 3.7222E-07 -2.3188E-08 1.0089E-09 R7 / / / / / / R8 / / / / / / R9 / / / / / / R10 / / / / / / R11 1.8275E+01 9.3322E-05 1.3877E-05 -5.9836E-08 3.1466E-11 -2.1374E-10 R12 -9.0000E+01 4.6139E-04 1.6877E-05 1.3448E-07 -3.5222E-09 1.0540E-10 R13 / / / / / / R14 / / / / / / R15 / / / / / / R16 / / / / / / conic coefficients aspheric coefficients k A14 A16 A18 A20 / R1 / / / / / / R2 / / / / / / R3 / / / / / / R4 / / / / / / R5 8.2388E+00 -3.5635E-11 3.4356E-13 -1.3188E-15 -2.2252E-15 / R6 -1.5012E-01 -4.1172E-12 -4.4906E-13 -1.7248E-14 9.7010E-16 / R7 / / / / / / R8 / / / / / / R9 / / / / / / R10 / / / / / / R11 1.8275E+01 2.2834E-12 7.0198E-14 -1.1661E-15 -8.0521E-17 / R12 -9.0000E+01 -4.7993E-12 -6.0343E-14 4.8322E-15 -6.9650E-17 / R13 / / / / / / R14 / / / / / / R15 / / / / / / R16 / / / / / /

[0179] Table 11 and Table 12 respectively show the design data of the inflection point and stationary point of each lens of the camera optical lens 30 in the third embodiment of the present disclosure.

TABLE-US-00011 TABLE 11 The number of inflection the location 1 of inflection points point P1R1 // P1R2 // P2R1 // P2R2 // P3R1 // P3R2 1 4.435 P4R1 // P4R2 // P5R1 // P5R2 // P6R1 // P6R2 // P7R1 // P7R2 // P8R1 // P8R2 //

TABLE-US-00012 TABLE 12 the number of stationary points P1R1 / P1R2 / P2R1 / P2R2 / P3R1 / P3R2 / P4R1 / P4R2 / P5R1 / P5R1 / P6R2 / P6R1 P7R1 / P7R2 / P8R1 / P8R2 /

[0180] Table 21, which appears later, shows the various values in the third embodiment, and the values corresponding with the parameters which are already fixed in the conditions.

[0181] FIG. 10 shows the schematic diagram of the field curvature and distortion after light with a wavelength of 555 nm passes through the camera optical lens 30 in the third embodiment. FIG. 11 shows the lateral color schematic diagram after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 30 in the third embodiment. FIG. 12 shows the longitudinal aberration after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 30 in the third embodiment.

[0182] As shown in Table 21, the third embodiment satisfies the various conditions.

[0183] In this embodiment, the pupil entering diameter (ENPD) of the camera optical lens 30 is 4.482 mm, the full vision field image height (IH) is 5.139 mm, and the vision field angle (FOV) in the diagonal direction is 72.88°. The camera optical lens 30 can meet the design requirements of large aperture and miniaturization, and chromatic aberrations on-axis and chromatic aberrations off-axis are adequately corrected. And the camera optical lens 30 has excellent optical characteristics.

Fourth Embodiment

[0184] FIG. 13 shows the camera optical lens 40 in the fourth embodiment of the present disclosure. The fourth embodiment is basically the same as the first embodiment, the meaning of its symbols is the same as that of the first embodiment, in the following, only the differences are described.

[0185] In this embodiment, the first lens of L1 has a convex image side surface at the proximal axis. The seventh lens L7 has a concave image side surface at the proximal axis. The eighth lens L8 has a convex image side surface at the proximal axis.

[0186] Table 13 and table 14 show the design data of the camera optical lens 40 in the fourth embodiment of the present disclosure.

TABLE-US-00013 TABLE 13 R d nd vd S1 ∞ d0= -21.942 R1 15.913 d1= 0.871 nd1 1.5891 vd1 61.25 R2 9.438 d2= 3.529 R3 55.161 d3= 0.922 nd2 1.7552 vd2 27.55 R4 221.022 d4= 16.037 R5 13.034 d5= 2.688 nd3 1.6192 vd3 63.85 R6 -12.369 d6= 0.199 R7 21.295 d7= 0.500 nd4 1.4970 vd4 81.59 R8 -3796.563 d8= 0.079 R9 -341.635 d9= 7.764 nd5 1.7618 vd5 26.61 R10 6.062 d10= 1.970 R11 25.554 d11= 2.710 nd6 1.8513 vd6 40.10 R12 -445.514 d12= 0.080 R13 9.087 d13= 1.010 nd7 1.7340 vd7 51.49 R14 22.081 d14= 1.077 R15 -19.032 d15= 0.921 nd8 2.1042 vd8 17.02 R16 -2545625.146 d16= 0.198 R17 ∞ d17= 0.900 ndg 1.5168 vgd 64.21 R18 ∞ d18= 0.199

TABLE-US-00014 TABLE 14 conic coefficients aspheric coefficients k A4 A6 A8 A10 A12 R1 / / / / / / R2 / / / / / / R3 / / / / / / R4 / / / / / / R5 5.3173E+00 -3.9655E-04 -1.3960E-05 7.1421E-07 -6.8632E-08 1.8423E-09 R6 -6.6883E-01 2.2442E-04 -5.4342E-06 2.9909E-07 -2.8813E-08 8.2019E-10 R7 / / / / / / R8 / / / / / / R9 / / / / / / R10 / / / / / / R11 1.1023E+01 1.6180E-03 1.1849E-05 -2.0244E-08 -1.3678E-08 -1.2432E-09 R12 -8.8142E+01 1.8916E-03 2.6446E-05 2.3877E-06 -2.9583E-08 -1.5009E-10 R13 / / / / / / R14 / / / / / / R15 / / / / / / R16 / / / / / / conic coefficients aspheric coefficients k A14 A16 A18 A20 / R1 / / / / / / R2 / / / / / / R3 / / / / / / R4 / / / / / / R5 5.3173E+00 -4.3182E-11 6.0836E-13 4.0757E-14 -2.6624E-15 / R6 -6.6883E-01 -5.0054E-12 -8.5748E-14 -6.2726E-15 -7.0490E-17 / R7 / / / / / / R8 / / / / / / R9 / / / / / / R10 / / / / / / R11 1.1023E+01 2.8152E-11 2.6030E-12 -2.9068E-16 -2.3510E-15 / R12 -8.8142E+01 9.1714E-11 -2.6529E-12 -6.7304E-13 3.5102E-14 / R13 / / / / / / R14 / / / / / / R15 / / / / / / R16 / / / / / /

[0187] Table 15 and Table 16 respectively show the design data of the inflection point and stationary point of each lens of the camera optical lens 40 in the fourth embodiment of the present disclosure.

TABLE-US-00015 TABLE 15 The number of inflection the location 1 of inflection points point P1R1 // P1R2 // P2R1 // P2R2 // P3R1 1 4.165 P3R2 // P4R1 // P4R2 // P5R1 // P5R2 // P6R1 // P6R2 1 0.315 P7R1 // P7R2 // P8R1 // P8R2 //

TABLE-US-00016 TABLE 16 the location 1 of stationary the number of stationary points point P1R1 // P1R2 // P2R1 // P2R2 // P3R1 // P3R2 // P4R1 // P4R2 // P5R1 // P5R2 // P6R1 // P6R2 1 0.545 P7R1 // P7R2 // P8R1 // P8R2 //

[0188] Table 21, which appears later, shows the various values in the fourth embodiment, and the values corresponding with the parameters which are already fixed in the conditions.

[0189] FIG. 14 shows the schematic diagram of the field curvature and distortion after light with a wavelength of 555 nm passes through the camera optical lens 40 in the fourth embodiment. FIG. 15 shows the lateral color schematic diagram after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 40 in the fourth embodiment. FIG. 16 shows the longitudinal aberration after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 40 in the fourth embodiment.

[0190] As shown in Table 21, the fourth embodiment satisfies the various conditions.

[0191] In this embodiment, the pupil entering diameter (ENPD) of the camera optical lens 40 is 6.915 mm, the full vision field image height (IH) is 5.139 mm, and the vision field angle (FOV) in the diagonal direction is 43.13°. The camera optical lens 40 can meet the design requirements of large aperture and miniaturization, and chromatic aberrations on-axis and chromatic aberrations off-axis are adequately corrected. And the camera optical lens 40 has excellent optical characteristics.

A Contrast Embodiment

[0192] FIG. 17 shows the camera optical lens 50 in the contrast embodiment. The meaning of symbols in the contrast embodiment is the same as the first embodiment. Only the differences are described below.

[0193] Table 17 and table 18 show the design data of the camera optical lens 50 in the contrast embodiment.

TABLE-US-00017 TABLE 17 R d nd vd S1 ∞ d0= -9.752 R1 15.235 d1= 1.300 nd1 1.5891 vd1 61.25 R2 6.090 d2= 0.665 R3 6.872 d3= 2.000 nd2 1.7552 vd2 27.55 R4 6.631 d4= 5.677 R5 14.259 d5= 1.919 nd3 1.6192 vd3 63.85 R6 -12.944 d6= 0.200 R7 -43.180 d7= 6.330 nd4 1.4970 vd4 81.59 R8 -6.956 d8= 0.050 R9 -7.516 d9= 0.250 nd5 1.7618 vd5 26.61 R10 -29.156 d10= 3.439 R11 20.835 d11= 1.381 nd6 1.8513 vd6 40.10 R12 599.901 d12= 1.450 R13 10.933 d13= 4.744 nd7 1.7340 vd7 51.49 R14 13.833 d14= 2.271 R15 -10.674 d15= 0.500 nd8 1.8467 vd8 23.79 R16 -123.659 d16= 0.200 R17 ∞ d17= 0.900 ndg 1.5168 vgd 64.21 R18 ∞ d18= 0.200

TABLE-US-00018 TABLE 18 conic coefficients aspheric coefficients k A4 A6 A8 A10 A12 R1 // // // // R2 // // // // R3 // // // // R4 // // // // R5 7.1097E+00 -7.4369E-04 1.5427E-05 -2.4480E-06 -6.1634E-08 6.1381E-09 R6 1.2180E+00 -1.2876E-04 -1.7511E-05 9.1781E-07 -6.1760E-08 -8.2329E-10 R7 // // // // R8 // // // // R9 // // // // R10 // // // // R11 9.2365E+00 4.9251E-04 4.6603E-06 6.8843E-08 4.9162E-10 -1.3981E-10 R12 8.2396E+01 7.6077E-04 1.3807E-05 -1.1166E-07 6.1183E-09 6.6475E-11 R13 // // // // R14 // // // // R15 // // // // R16 // // // // conic coefficients aspheric coefficients k A14 A16 A18 A20 / R1 // // // // R2 // // // // R3 // // // // R4 // // // // R5 7.1097E+00 2.1801E-11 -6.4105E-12 -3.6243E-13 4.1102E-16 / R6 1.2180E+00 2.0855E-11 3.9820E-12 1.8157E-14 -1.4529E-14 / R7 // // // // R8 // // // // R9 // // // // R10 // // // // R11 9.2365E+00 4.3988E-12 4.2025E-15 -1.8140E-15 4.5558E-17 / R12 8.2396E+01 -1.0447E-12 -1.3425E-13 3.9939E-16 1.7040E-16 / R13 // // // // R14 // // // // R15 // // // // R16 // // // //

[0194] Table 19 and Table 20 respectively show the design data of the inflection point and stationary point of each lens of the camera optical lens 50 in the contrast embodiment of the present disclosure.

TABLE-US-00019 TABLE 19 The number of inflection the location 1 of inflection points point P1R1 // P1R2 // P2R1 // P2R2 // / P3R1 1 2.965 P3R2 // P4R1 // P4R2 // P5R1 // P5R2 // P6R1 // P6R2 // P7R1 // P7R2 // P8R1 // P8R2 //

TABLE-US-00020 TABLE 20 the number of stationary points the location of stationary point P1R1 // P1R2 // P2R1 // P2R2 // / P3R1 1 3.995 P3R2 // P4R1 // P4R2 // P5R1 // P5R1 // P6R2 // P6R1 // P7R1 // P7R2 // P8R1 // P8R2 //

[0195] FIG. 18 shows the schematic diagram of the field curvature and distortion after light with a wavelength of 555 nm passes through the camera optical lens 50 in the contrast embodiment. FIG. 19 shows the lateral color schematic diagram after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 50 in the contrast embodiment. FIG. 20 shows the longitudinal aberration after light with wavelengths of 470 nm, 510 nm, 555 nm, 610 nm and 650 nm passes through the camera optical lens 50 in the contrast embodiment.

[0196] Table 21 below shows the various values in this embodiment in accordance with the above conditions. Obviously, the camera optical lens 50 in this embodiment does not satisfy the above condition: $0.20 \leq R3/R4 \leq 0.90$.

[0197] In the contrast embodiment, the pupil entering diameter (ENPD) of the camera optical lens 50 is 6.283 mm, the full vision field image height (IH) is 5.139 mm, and the vision field angle (FOV) in the diagonal direction is 54.89°. The camera optical lens 50 does not meet the design requirements of large aperture, wide-angle and ultra-thin, and chromatic aberrations on-axis and chromatic aberrations off-axis are not adequately corrected.

TABLE-US-00021 TABLE 21 Parameters and The first The second The third The fourth the contrast conditions Embodiment Embodiment Embodiment embodiment d3/d1 1.488 2.996 1.011 1.059 1.538 R3/R4 0.479 0.205 0.890 0.250 1.036 f4/f3 1.908 1.441 1.007 3.991 1.401 f6/f7 0.960 0.506 0.520 1.396 0.603 nd8 1.847 1.847 1.805 2.104 1.847 |f8/f| 1.551 3.991 1.751 1.501 1.329 (FOV*f)/IH 122.960 116.021 112.838 95.345 110.256 f 5.937 7.681 7.956 11.361 10.324 f1 -7.404 -10.306 -13.273 -41.319 -18.127 f2 64.443 48.844 91.369 96.468 96.288 f3 9.761 10.472 12.042 10.652 11.229 f4 18.384 15.092 12.126 42.515 15.733 f5 -10.328 -9.189 -10.304 -7.691 -13.268 f6 11.93 26.219 30.016 28.335 25.211 f7 12.262 51.773 57.733 20.296 41.8 f8 -9.578 -30.659 -13.934 -17.058 -13.723 f12 -8.547 -13.774 -15.291 -75.731 -19.263 FNO 1.64 1.64 1.64 1.64 1.64

[0198] The above embodiments of the present disclosure provide a detailed description of the camera optical lens. Specific examples are applied herein to illustrate the principles and embodiments of the present disclosure. The above description of the embodiments is only used to help understand the spirit and scope of present disclosure, various modifications may be made to these embodiments in terms of formality and detail, without departing from the spirit and scope of the present disclosure. In summary, the contents of the present disclosure should not be construed as a limitation of present disclosure.

Claims

1. A camera optical lens, comprising, from an object side to an image side in sequence: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; a seventh lens; a eighth lens; wherein the camera optical lens satisfies the following

conditions;

$1.00 \leq d3/d1 \leq 3.00$;

$0.20 \leq R3/R4 \leq 0.90$;

$1.00 \leq f4/f3 \leq 4.00$;

$0.50 \leq f6/f7 \leq 1.40$;

$1.80 \leq nd8 \leq 2.20$; where, d1 represents a thickness on-axis of the first lens; d3 represents a thickness on-axis of the second lens; R3 represents a curvature radius of the object side surface of the second lens; R4 represents a curvature radius of the image side surface of the second lens; f3 represents a focal length of the third lens; f4 represents a focal length of the fourth lens; f6 represents a focal length of the sixth lens; f7 represents a focal length of the seventh lens; nd8 represents a refractive index of the eighth lens.

2. The camera optical lens according to claim 1, wherein the camera optical lens further satisfies the following conditions:

$|f8/f| \leq 3.00$; where, f8 represents a focal length of the eighth lens; f represents a focal length of the camera optical lens.

3. The camera optical lens according to claim 1, wherein the camera optical lens has a field of view; and the camera optical lens further satisfies the following conditions: $(FOV * f) / IH \geq 95$; where, FOV represents the field of view of the camera optical lens; f represents a focal length of the camera optical lens; IH represents an image height of the camera optical lens.

4. The camera optical lens according to claim 3, wherein the camera optical lens further satisfies the following conditions: $(FOV * f) / IH \leq 1.25$.

5. The camera optical lens according to claim 1, wherein the first lens has a negative refractive power, and the first lens has a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions:

$0.44 \leq (R1 + R2) / (R1 - R2) \leq 5.87$; $-7.27 \leq f1/f \leq -0.83$; $0.01 \leq d1/TTL \leq 0.05$; where, R1 represents a curvature radius of the object side surface of the first lens; R2 represents a curvature radius of the image side surface of the first lens; f1 represents a focal length of the first lens; f represents a focal length of the camera optical lens; TTL represents a total optical length of the camera optical lens.

6. The camera optical lens according to claim 1, wherein the second lens has a positive refractive power, and the second lens has a convex object side surface at a proximal axis and a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-34.47 \leq (R3 + R4) / (R3 - R4) \leq -1.01$; $3.18 \leq f2/f \leq 17.23$;

$0.01 \leq d3/TTL \leq 0.08$; where, f2 represents a focal length of the second lens; f represents a focal length of the camera optical lens; TTL represents a total optical length of the camera optical lens.

7. The camera optical lens according to claim 1, wherein the third lens has a positive refractive power, and the third lens has a convex object side surface at a proximal axis and a convex image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $0.01 \leq (R5 + R6) / (R5 - R6) \leq 0.62$; $0.47 \leq f3/f \leq 2.47$; $0.03 \leq d5/TTL \leq 0.22$; where, R5 represents a curvature radius of the object side surface of the third lens; R6 represents a curvature radius of the image side surface of the third lens; f represents a focal length of the camera optical lens; d5 represents a thickness on-axis of the third lens; TTL represents a total optical length of the camera optical lens.

8. The camera optical lens according to claim 1, wherein the fourth lens has a positive refractive power, and the fourth lens has a convex object side surface at a proximal axis and a convex image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-1.98 \leq (R7 + R8) / (R7 - R8) \leq 1.12$; $0.76 \leq f4/f \leq 5.61$; $0.01 \leq d7/TTL \leq 0.23$; where, R7 represents a curvature radius of the object side surface of the fourth lens; R8 represents a curvature radius of the image side surface of the fourth lens; f represents a focal length of the camera optical lens; d7 represents a thickness on-axis of the fourth lens; TTL represents a total optical length of the camera optical lens.

9. The camera optical lens according to claim 1, wherein the fifth lens has a negative refractive power, and the fifth lens has a concave object side surface at a proximal axis and a concave image side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-0.99 \leq (R9 + R10) / (R9 - R10) \leq 1.45$; $-3.48 \leq f5/f \leq -0.45$; $0.00 \leq d9/TTL \leq 0.28$; where, R9 represents a curvature radius of the object side surface of the fifth lens; R10 represents a curvature radius of the image side surface of the fifth lens; f represents a focal length of the camera optical lens; d9 represents a thickness on-axis of the fifth lens; TTL represents a total optical length of the camera optical lens.

10. The camera optical lens according to claim 1, wherein the sixth lens has a positive refractive power, and the sixth lens has a convex object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions: $-2.00 \leq (R11 + R12) / (R11 - R12) \leq 0.22$; $1.00 \leq f6/f \leq 5.66$; $0.01 \leq d11/TTL \leq 0.14$; where, R11 represents a curvature radius of the object side surface of the sixth lens; R12 represents a curvature radius of the image side surface of the sixth lens; f represents a focal length of the camera optical lens; d11 represents a thickness on-axis of the sixth lens; TTL represents a total optical length of the camera optical lens.

11. The camera optical lens according to claim 1, wherein the seventh lens has a positive refractive power, and the seventh lens has a convex object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions:

$-11.16 \leq (R13 + R14) / (R13 - R14) \leq -0.01$; $0.89 \leq f7/f \leq 10.88$; $0.01 \leq d13/TTL \leq 0.30$; where, R13 represents a curvature radius of the object side surface of the seventh lens; R14 represents a curvature radius of the image side surface of the seventh lens; f represents a focal length of the camera optical lens; d13 represents a thickness on-axis of the seventh lens; TTL represents a total optical length of the camera optical lens.

12. The camera optical lens according to claim 1, wherein the eighth lens has a negative refractive power, and the eighth lens has a concave object side surface at a proximal axis; wherein the camera optical lens further satisfies the following conditions:

$-12.50 \leq (R15 + R16) / (R15 - R16) \leq 0.02$; $0.01 \leq d15/TTL \leq 0.05$; where, R15 represents a curvature radius of the object side surface of the eighth lens; R16 represents a curvature radius of the image side surface of the eighth lens; d15 represents a thickness on-axis of the eighth lens; TTL represents a total optical length of the camera optical lens.

13. The camera optical lens according to claim 1, wherein the first lens is made of glass material.

14. The camera optical lens according to claim 1, wherein the second lens is made of glass material.

15. The camera optical lens according to claim 1, wherein the third lens is made of glass material.

- 16.** The camera optical lens according to claim 1, wherein the fourth lens is made of glass material.
 - 17.** The camera optical lens according to claim 1, wherein the fifth lens is made of glass material.
 - 18.** The camera optical lens according to claim 1, wherein the sixth lens is made of glass material.
 - 19.** The camera optical lens according to claim 1, wherein the seventh lens is made of glass material.
 - 20.** The camera optical lens according to claim 1, wherein the eighth lens is made of glass material.
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